



INT 332 – Devops Virtualization And Configuration Management

Introduction to GitHub Actions



GitHub Actions is a CI/CD (Continuous Integration & Continuous Deployment) platform that allows you to automate workflows directly inside your GitHub repository.

It helps in:

- **Building code**
- **Running tests**
- **Deploying applications**
- **Automating repetitive development tasks**

Analogy:

Think of GitHub Actions like a smart assistant that automatically performs tasks (like testing or deployment) whenever something happens in your repository.



Understanding Workflow Automation



What is a Workflow?

- A workflow is an automated process defined in a YAML file.
- Stored inside `.github/workflows/`
- Contains one or more jobs
- Each job has steps (commands/actions)

Basic Structure of a Workflow

name: My First Workflow

on: push

jobs:

build:

runs-on: ubuntu-latest

steps:

- **name:** Checkout code

uses: actions/checkout@v3

- **name:** Run a script

run: echo "Hello World"

Components of Workflow



- **Event (on) → When to run**
- **Jobs → What to do**
- **Steps → How to do**
- **Actions → Predefined reusable tasks**

Events in GitHub Actions

What are Events?

- **Events are activities that trigger workflows.**

Event	Description
push	When code is pushed
pull_request	When PR is created/updated
workflow_dispatch	Manual trigger
schedule	Runs on cron timing
issues	When issue is created/updated

Triggers in GitHub Actions

What is a Trigger?

A trigger is the condition under which an event runs a workflow.

- Event = "push"
Trigger = "only on main branch"

Example with Filters

on:

push:

branches:

- main

paths:

- - 'src/**'

Types of Triggers

1. Branch-based Trigger

branches: [main, dev]

2. Path-based Trigger

paths: - 'app/**'

3. Tag-based Trigger

tags:- 'v1.*'



4. Time-based Trigger (Cron)

on:

schedule:

- cron: '0 0 * * *'

5. Workflow Directory Structure

repository/

```
|— .github/
|   |— workflows/
|       |— build.yml
|       |— deploy.yml
|
|— src/
|— tests/
|— README.md
```



6. Jobs and Steps

Jobs

A workflow contains multiple jobs

Jobs run in parallel by default

jobs:

build:

test:



Steps

- Each job has steps executed sequentially

steps:

- name: Step 1

run: echo "Hello"

7. Actions (Reusable Components)

What are Actions?

Reusable units of code.

Example:

uses: actions/checkout@v3

8. Runners

What is a Runner?

A server that runs your workflow.

Types:

- **GitHub-hosted (Ubuntu, Windows, macOS)**
- **Self-hosted**

Real-Life Example Workflow



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name: Java CI Pipeline

- **on:**
- **push:**
- **branches: [main]**
- **jobs:**
- **build:**
- **runs-on: ubuntu-latest**
- **steps:**
 - **- uses: actions/checkout@v3**
 - **- name: Set up Java**
 - **uses: actions/setup-java@v3**
 - **with:**
 - **java-version: '17'**
 - **- name: Build with Maven**
 - **run: mvn clean install**



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GitHub Actions Workflows

1. Workflow Triggers in GitHub Actions



- **1.1 Push Trigger**

Runs workflow when code is pushed.

on:

push:

branches: [main]

Use case:

- **Continuous Integration (CI)**
- **Build & test on every commit**

1.2 Pull Request Trigger



- Triggered when a PR is opened, updated, or merged.

on:

pull_request:

branches: [main]

Use case:

- Code review validation
- Testing before merging

1.3 Schedule Trigger (Cron Jobs)

Runs workflow at scheduled intervals.

on:

schedule:

- cron: '0 0 * * *'

Example:

- **Nightly builds**
- **Automated backups**

1.4 Manual Trigger (workflow_dispatch)

Allows user to trigger workflow manually from GitHub UI.

on:

workflow_dispatch:

Use case:

- **On-demand deployments**
- **Testing pipelines manually**

2.Jobs & Matrix Strategies

What is a Job?

A job is a set of steps executed on the same runner.

jobs:

build:

runs-on: ubuntu-latest

Matrix Strategy (Parallel Execution)

- Runs the same job with multiple configurations.

jobs:

test:

runs-on: ubuntu-latest

strategy:

matrix:

node-version: [14, 16, 18]

steps:

- uses: actions/setup-node@v3

with:

node-version: \${matrix.node-version}

3. Steps & Shell Commands

Steps

- Each job consists of sequential steps.

Steps:

- - name: Step 1
- run: echo "Hello World"

Shell Commands



- **Run OS-level commands directly.**

- name: List files

run: ls -la

Quick Quiz



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1. Identify the correct order of execution:

- **A. Steps → Jobs → Workflow**
- B. Workflow → Jobs → Steps**
- C. Jobs → Workflow → Steps**
- D. Steps → Workflow → Jobs**

2. Which of the following is a reusable unit in GitHub Actions?

- A. Job**
- B. Step**
- C. Action**
- D. Runner**



3. A workflow in GitHub Actions is defined using:

- **A. JSON file**
- B. YAML file**
- C. XML file**
- D. Properties file**

4. Where are workflow files stored in a repository?

- **A. /src/workflows/**
- B. /github/actions/**
- C. /.github/workflows/**
- D. /ci-cd/**



5. Which component defines *when* a workflow runs?

- A. Jobs
- B. Steps
- C. Events
- D. Actions



6. Which of the following is a valid push trigger?

A.

- **on: push**

B.

- **on:
 push:
 branches: [main]**

C.

- **trigger: push**

D.

- **event: push**



7. What is the role of a runner?

- A. Stores code**
- B. Executes workflows**
- C. Triggers events**
- D. Builds UI**

8. Jobs in a workflow run:

- A. Sequentially by default**
- B. In parallel by default**
- C. Only manually**
- D. Only on Windows**





9. Steps inside a job run:

- **A. In parallel**
B. Randomly
C. Sequentially
D. Only once



10. Identify the correct order of execution:

- A. Steps → Jobs → Workflow**
- B. Workflow → Jobs → Steps**
- C. Jobs → Workflow → Steps**
- D. Steps → Workflow → Jobs**



11. What does this cron expression mean?

cron: '0 0 * * *'

- A. Every hour**
- B. Every minute**
- C. Every day at midnight**
- D. Every week**



12. You want to test code on multiple Node.js versions simultaneously. What should you use?

- A. Steps**
- B. Matrix strategy**
- C. Runner**
- D. Trigger**



13. If a workflow fails at step 2, what happens?

- A. Step 3 still runs**
- B. Job stops execution**
- C. Workflow restarts**
- D. Next job starts**



14. GitHub-hosted runners support:

- A. Only Linux
- B. Only Windows
- C. Linux, Windows, macOS
- D. None

Practice Questions:

1. Identify Error

- Find the error in the following YAML:

name: CI Pipeline

on push:

branches:

- main

jobs:

build:

runs-on ubuntu-latest

steps:

- name Checkout code

uses actions/checkout@v3

2.

Workflow Trigger

➤ Write a GitHub Actions workflow that triggers:

On push

Only for develop branch

3. Basic Workflow File



➤ Create a minimal GitHub Actions .yml file that:

Checks out code

Prints "Hello World"

4. Complete CI Workflow

- Write a full .yml file that:
 - Triggers on push
 - Runs on Ubuntu
 - Steps:
 - Checkout code
 - Build project (use echo "Building")
 - Test project (use echo "Testing")

5. Multiple Steps

➤ Write a job with two steps:

- Install dependencies
- Run tests

6.



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jobs:

build:

runs-on: ubuntu-latest

steps:

- name: Build

run: ls

7.

- **name: Node CI**
- **on:**
 - **push:**
 - **branches:**
 - **- main**
- **jobs:**
 - **build:**
 - **runs-on: ubuntu-latest**
 - **steps:**
 - **- name: Install dependencies**
 - **run: npm install**
 - **- name: Run tests**
 - **run: npm test**



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